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Title of Article

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Key Points:

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- RSTF

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- Source Inversion using RSTF and Teleseismic data

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- Compare slip models from RSTF and Teleseismic data

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Abstract

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Plain Language Summary

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1 Introduction**1.1 Elastic Rebound Theory and Earthquake Cycles on Transform Faults**

- In simple cases of stress accumulation and release of seismic faults, earthquakes are expected to occur quasi-periodically, with the recurrence interval determined by the rate of tectonic loading and the amount of slip released during each earthquakes (Rong et al., 2003).
- The elastic rebound theory (Reid, 1910), together with the seismic gap hypothesis, provides a conceptual framework for understanding the accumulation and subsequent release of tectonic strain along fault boundaries.
- Shortcomings of the elastic rebound theory in explaining complex earthquake cycles, especially on oceanic transform faults : It neglects the fact that the rupture process may not be uniform along the fault plane, and that the slip distribution can be highly heterogeneous (Heaton, 1990).

1.2 Seismotectonic Setting of the Study Area and Fault Structure

- The study area lies along an active plate boundary characterized by quasi-periodic seismicity (Aderhold & Abercrombie, 2016). The fault has an approximate strike of $98/75/175^\circ$ and accommodates relative plate motion at an average slip rate of approximately 2.24 cm/yr (Aderhold & Abercrombie, 2016; DeMets et al., 1990).
- The fault system has traditionally been divided into two major segments, Segment A and Segment B (Searle, 1981; Whitmarsh & Calvert, 1986; Skolotnev et al., 2021). Segment A, in particular, shows significant seismicity (Aderhold & Abercrombie, 2016).
- Recent observations of earthquake mechanisms and fault geometry suggest that Segment A can be further divided into two structurally distinct portions: Segment A-East, characterized predominantly by near-vertical strike-slip motion, and Segment A-West, where the fault dips northward and exhibits a component of transpression (Janin et al., 2026).
- Our particular interest are the 2015 (M_w 7.1) and 2025 (M_w 6.9) earthquake sequences, which occurred along the same fault system within a ten-year interval. Their close temporal and spatial association provides an opportunity to investigate earthquake recurrence, rupture segmentation, and variations in slip distribution along the fault system (Figure 1a).

1.3 Unresolved Questions and Study Objectives

- As shown in Figure 1, the 2025 earthquake occurred only about 10 years after the 2015 earthquake and affected the same section of the Charlie–Gibbs transform fault. Such a short recurrence interval is not evident in the historical seismicity of the fault (Figure 1b). Given the relatively slow plate motion of ~ 2.24 cm/yr (DeMets et al., 1990), only about 22 cm of tectonic displacement would accumulate over 10 years. This is much smaller than the several metres of slip typically associated with a large earthquake. The occurrence of two large earthquakes on the same fault segment within such a short time interval is therefore unexpected in the frame-

- work of a simple stress-loading and elastic-rebound model, where a major earthquake is expected to release accumulated strain and consequently delay another large rupture on the same segment.
- To investigate: Are the slip distributions of the 2015 and 2025 earthquakes similar or different? If different, what are the implications for stress accumulation and release along the fault system?

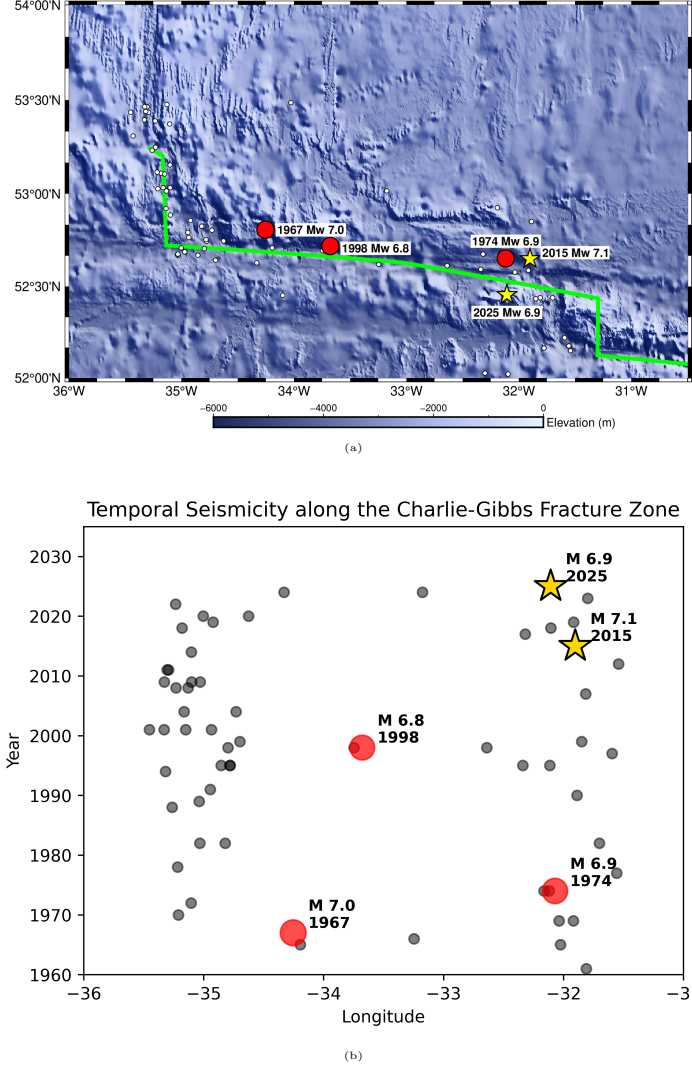


Figure 1: Seismicity and tectonic setting of the northern Charlie-Gibbs transform fault. (a) Regional tectonic setting shown on a global bathymetric map derived from the GMT Earth Relief dataset. (b) Detailed bathymetric map of the northern Charlie-Gibbs transform fault, based on the 3-arc-sec GMT Earth Relief grid (SRTM3S), with the fault trace shown as a solid line referring from Aderhold et al. (2016). Earthquake epicenters are from USGS ComCat, with magnitudes compiled from USGS ComCat and historical earthquake catalogues where available. Circle sizes scale with earthquake magnitude. The 1967 (M_w 7.0), 1974 (M_w 6.9), and 1998 (M_w 6.8) earthquakes are shown as colored circles, while the 2015 (M_w 7.1) and 2025 (M_w 6.9) earthquakes are shown as stars. (c) Temporal distribution of earthquakes as a function of longitude, illustrating the spatial and temporal recurrence of major earthquakes along the Charlie-Gibbs transform fault. Bathymetric data are from GMT Earth Relief; the 15-arc-sec regional grid is based on SRTM15+ V2.1 and the 3-arc-sec detailed grid is based on SRTM3S.

2 Data and Methodology

2.1 Data

- Continuous three-component waveform time series and their associated metadata for both the 2015 and 2025 events were retrieved from GEOSCOPE and other broadband databases, covering networks IU, II, GT, G, GE, DK, and CN via the `FetchData` utility.
- We selected P - and SH -wave recordings from stations located within an epicentral distance range of 30° to 80° for both earthquakes. The selected array provides well-distributed azimuthal coverage in each case; their spatial distribution is illustrated in Figure S1a (for the 2015 event) and Figure S1b (for the 2025 event).
- A bandpass filter of 0.005 to 0.2 Hz was applied to the obtained data to remove high-frequency noise and low-frequency trends and a combination of automated and manual phase picking was performed to identify the arrival times of the P and SH waves. The picked phases were then used to window the data for subsequent analysis.
- For RSTF deconvolution, we retrieved data for a radius of 40 degrees around the epicentres of the 2015 and 2025 earthquakes for same set of networks as we used for teleseismic data but at regional to teleseismic distances. We had good azimuthal coverage of the data for both earthquakes. Their spatial distribution is illustrated in Figure S1c (for the 2015 event) and Figure S1d (for the 2025 event).
- For RSTF analysis, we selected the same earthquake which happened on 05/04/2025 as our EGF for both 2015 and 2025 events. To qualify as a good EGF, the earthquake should be smaller, have a similar focal mechanism and be located close to the mainshock. This mentioned earthquake is selected as EGF because it satisfies all the mentioned criteria.
- The retrieved data were preprocessed by removing the instrument response and applying a bandpass filter between 0.0125 and 0.02 Hz. After that Love-wave windows were picked.
- Then Relative Source Time Functions (RSTFs) were calculated via the deconvolution method of (Vallée, 2004) which will be served as input data for the finite-fault inversion.
- The RSTFs (Figure ?? and Figure ??) show a clear azimuthal dependence, with a stretched profile toward the east and a compressed profile toward the west. This pattern is consistent with the directivity effect caused by rupture propagation from west to east (Aderhold & Abercrombie, 2016). The same pattern is observed for both the 2015 and 2025 earthquakes, suggesting that the two events had a similar rupture propagation direction.
- However, the RSTFs for the 2025 earthquake show stronger stretching and compression compared to those of the 2015 earthquake. This suggests that the 2025 rupture was more complex and lasted longer than the 2015 rupture.

2.2 Methodology

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2.3 Finite-fault kinematic inversion

Fault Plane Geometry

For both the 2015 and 2025 events, we adopted a consistent fault geometry and an 8×8 km subfault discretization, following the geometric constraints established by (Aderhold & Abercrombie, 2016). Given their very similar focal depths, the hypocenters for both earthquakes were fixed at the same spatial location—specifically at subfault 71. The defined fault plane spans 200 km in length and 32 km in width, with a strike of 98° , an ini-

tial dip of 75° , and an initial rake of 175° . To accommodate potential variations along the rupture surface during inversion, both rake and dip were permitted to deviate by up to $\pm 30^\circ$.

Inversion Configuration

As initial configuration for the inversion, we set seismic moment of 5.6×10^{26} dyn · cm for the 2015 earthquake and 3×10^{26} Nm for the 2025 earthquake, based on preliminary estimates from Mw 7.1. and Mw 6.9, respectively. The triangular source-time function for the subfaults is parameterized by 1 time window with a half-duration of 1.0 s, evaluated with a medium rigidity (μ) of $4e11$ dyn/cm², an initial synthetic triangle amplitude of $5e23$ dyn·cm, a maximum allowable time delay of 30 s for 2015 and 60 s for 2025, a subfault point-source grid spacing (Δx) of 2 km, and a local rupture propagation velocity of 3.5 km/s. Relative dataset weighting allocates 1.0 to teleseismic records and 1.0 to relative source-time functions (RSTFs), while assigning 0.0 to strong-motion, InSAR, GPS, and tsunami data as non-contributing. Inversion controls and regularization specify a moment minimization coefficient of 0.1, rupture velocity bounds from 2.0 km/s to 5.0 km/s, and spatial smoothing coefficients of 0.01 for slip, 0.0 for rupture velocity, 0.025 for rake variations, and 0.080 for dip variations.

This inversion solves for the spatial and temporal characteristics of the rupture. The principal source parameters include: spatial distribution of slip amplitude, rake, spatial variation in fault dip, source time function (STF) of the earthquake and rupture velocity.

2.3.1 Comparison of the 2015 and 2025 earthquake slip distributions

- Slip distributions are plotted for both the 2015 and 2025 earthquakes. The slip distributions are compared to understand the similarities and differences in the rupture processes of the two events.
- Contours of the slips from both earthquakes are plotted in a single figure to visualize the spatial relationship between the two slip distributions.
- To plot the contours we distributed the absolute slip values into 5 equal intervals starting from minimum slip of 8cm and plotted the contours for both earthquakes in a single figure.

2.3.2 Comparison between source time functions of the 2015 and 2025 earthquakes

- The STFs are plotted for both earthquakes and the differences in the duration, peak amplitude, and overall shape of the STFs are analyzed.

Results

2.4 Joint inversion results

- The fault slip distribution inversion is based on the analysis of the worldwide broadband teleseismic P- and S-wave data shown in Figure 1c and Figure 1d.
- Using the same fault geometry for both earthquakes, we inverted for the fault slip distribution. The resulting models show good agreement between the observed and predicted teleseismic waveforms for both events, as shown in Figure 2b and Figure 2d. The aggregate RMS misfit is 0.394 for the 2015 earthquake and 0.453 for the 2025 earthquake.

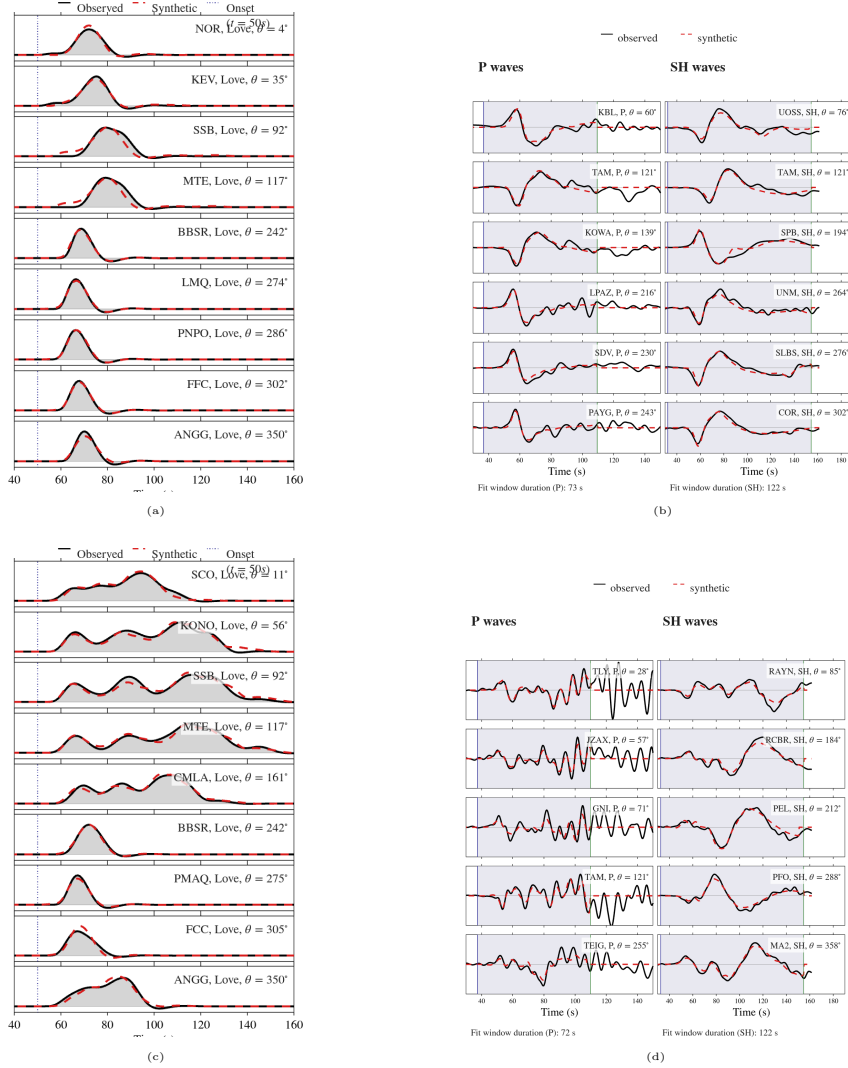


Figure 2: Joint inversion agreement with observed data (a) Observed (black) and synthetic (red) RSTFs for the 2015 earthquake, arranged according to station azimuth. Station names and azimuths (θ) are indicated with the corresponding RSTFs. (b) Same as (a) for the 2025 earthquake. The comparison between the observed and synthetic RSTFs shows the fit from the inversion of the fault slip distribution, with an aggregate RMS misfit of 0.128 for the 2015 earthquake and 0.126 for the 2025 earthquake. (c) Comparison between observed (black) and synthetic (red) teleseismic waveforms for the 2015 earthquake, arranged according to station azimuth. Station names and azimuths (θ) are indicated with the corresponding waveforms. (d) Same as (c) for the 2025 earthquake. The comparison between the observed and synthetic

- And the RSTFs obtained from the inverted slip distribution also show good agreement too with the observed RSTFs, as shown in Figure ?? having an aggregate rms of 0.126 for 2025 and 0.128 for 2015.

- The inverted slip distributions for the 2015 and 2025 earthquakes are shown in Figure ?? and Figure ??, respectively. The 2015 earthquake exhibits a concentrated slip distribution with a maximum slip of 1.1 m, while the 2025 earthquake shows a more distributed slip pattern with a maximum slip of 0.27 m. The total seismic moment released by the 2015 earthquake is 4.16×10^{19} Nm, whereas the 2025 earthquake released 2.89×10^{19} Nm. Our inversion found the average rupture velocity of around 2.7 km/s for 2015 which is consistent with the findings of 2-4 km/s (Aderhold and Abercrombie (2016)), and 3.1 km/s for 2025. These results indicate that the two earthquakes had different rupture characteristics, with the 2025 event being more spatially extended and having a longer duration than the 2015 event.
- The comparison of the retrieved source time functions shows the differences in the rupture duration and moment release between the two earthquakes. The 2015 earthquake had a rupture duration of about 35 seconds with a strong peak in moment rate. In contrast the 2025 earthquake released moment at a lower amplitude but over a longer period lasting around 60 seconds. These patterns highlight rupture behaviors, between the two events. STF's are shown in Figure ??.
- The rake distribution for both earthquakes is shown in Figure ?? and Figure ??. Both earthquakes show similar rake patterns towards west direction with average rake amount of 175° and 184° for the 2015 and 2025 earthquakes, respectively.
- The 2015 earthquake shows a relatively concentrated slip distribution, with the main asperity confined to the 0- -60 km (Figure ??) section along strike. In contrast, the 2025 earthquake exhibits a more distributed slip pattern with multiple asperities. One asperity is located around -20 to +20 km along strike, a second extends from approximately -30 to -60 km, while a third occupies the western section of the fault, extending from approximately -70 to -148 km (Figure ??).
- To investigate the relationship between the slip distributions of the two earthquakes, we compared the slip contours of both events in a single figure (Figure 3). It seen observed that in the region where most of the 2015 slip occurred, the 2025 earthquake shows low to no slip suggesting that the 2015 slip asperity may have released the accumulated stress in that region which did not have enough time to re-accumulate before the 2025 earthquake occurred. But there are also some overlapping regions which can be result of dynamic stress transfer from 2015 earthquake. But it was out of the scope of this study to investigate the stress transfer and stress accumulation along the fault system but can be investigated in future studies.
- We also tried to model dip as we kept dip as a free parameter in the inversion and this improved our model fit to the data. Inversion results from fixed dip is shown in the supplementary material.

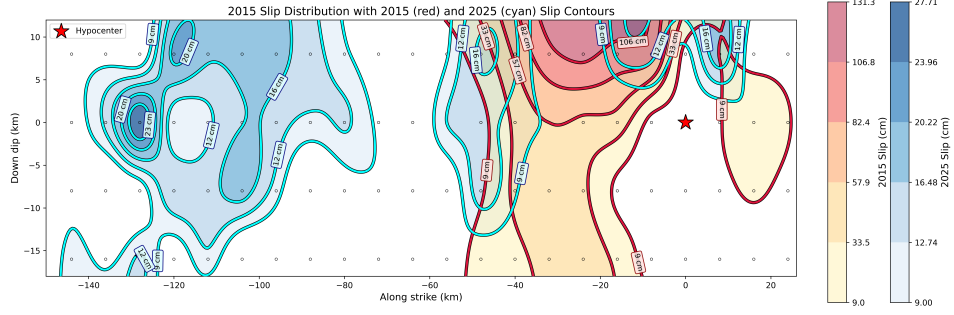


Figure 3: Comparison of the inverted slip distributions for the 2015 and 2025 Charlie-Gibbs earthquakes on the same fault geometry. The color-filled distribution represents the 2015 slip amplitude, with red contours showing the 2015 slip distribution and cyan contours showing the 2025 slip distribution. Red and cyan stars mark the hypocenters of the 2015 and 2025 earthquakes, respectively. The horizontal and vertical axes represent distance along strike and down dip, respectively.

3 Discussion and Conclusions

- Earthquake sequences involving partially overlapping rupture zones have been reported for several strike-slip fault systems, with successive large events separated by years to decades. Examples include the 1940 (M_w 7.1) and 1979 (M_w 6.6) earthquakes in the Imperial Valley (Rockwell & Klinger, 2013), the 1981 (M_w 7.1) and 1998 (M_w 6.6) earthquakes on the Gowk fault, Iran (Walker et al., 2010), and the 1979 (M_w 6.6), 1997 (M_w 7.2) Zirkuh earthquakes (Berberian et al., 1999) and earthquake sequences along the North Anatolian Fault (Barka, 1996). The recurrence of slip on previously ruptured sections were explained by spatial variations in co-seismic slip, whereas areas experiencing relatively limited slip during one earthquake can become sites of renewed rupture in a subsequent event. In other cases, the elapsed time between earthquakes was sufficient for tectonic loading to contribute substantially to the stress state before the subsequent rupture (King & Thatcher, 1998). In contrast with the above examples, in terms of magnitude 2015 and 2025 should mark the end of a sequence of earthquake cycle on the eastern segment which was previously seen in between 1974 (M_w 6.9) and 2015 (M_w 7.1) events.
- How such a short gap between these two large events on the same fault segment can be explained? The main clue lies in the source slip distribution of both the earthquakes. 2015 earthquake had one single confined asperity in the eastern segment of the fault with a maximum slip of 1.1m in the centre releasing the accumulated stress. In the other hand, 2025 earthquake had multiple small asperities with maximum slip of only 0.3m. From the seismic moment release point of view, to create such a large earthquake(2025) with such small slip amount, the rupture area must be very large which can be seen in the slip distribution (Figure ??)
- Shear cracks with finite cohesive forces can propagate by skipping barriers, which either persist or rupture under subsequent dynamic stress (Heaton, 1990) depending on the barrier strength-to-tectonic stress ratio (Das & Aki, 1977). In this context, the 2015 earthquake likely left unbroken barriers and generated new ones through stress redistribution. Because the dynamic stress increase at the rupture front can exceed the static stress modifications induced by the 2015 event, the 2025 rupture was able to break through residual barriers while skipping newly formed ones, resulting in multiple small slip asperities.
- The colocation of earthquakes on the same fault system within only ten years gives clear instrumental evidence that goes against the idea that the ground stops shaking after a rupture. This is similar to records that show clusters of events with time gaps between them that are much shorter than the average cycle. The 2015 and 2025 sequence shows that fault segments that have already broken can

break again in than a decade. Whether this happens because of a dynamic stress drop, multimodal slip or uneven loading of asperities the short-interval reactivation has a direct effect on how we assess seismic hazard. A recent large earthquake— one that is close to the biggest expected magnitude—does not make a fault segment safe from a near-future rupture. Instead new events that start in unruptured gaps or, on adjacent fault strands can spread into the area that was just ruptured.

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